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RESEARCH

Benefits of Niche Construction in Northern Rio Grande Pueblo Agriculture: Modeling Cotton, Maize, and Wheat Cultivation

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Abstract
To understand agricultural intensification as a form of niche construction, researchers need to be able to distinguish passive agricultural niches, where agronomic parameters were met by environmental conditions, from spaces where societies had to exert additional effort to augment the landscape to enable crop growth. The authors present an agricultural potential model that identifies passive agricultural niches and niches that required additional constructed effort and assesses the differential success of these areas for different crops by generating yield estimates. These estimates are then situated in cultural context of population, redistribution requirements, *etc*. to assess the sustainability of the system. The model is demonstrated through a case study in the Northern Rio Grande region of New Mexico, focusing on maize, cotton, and wheat, important crops for Ancestral Pueblo people. We use the PaleoCAR framework to assess locations where growing requirements for maize, cotton, and wheat were met, and what estimated yields (productivity) could have been in each agricultural catchment relative to the associated village population of four case study villages. The model suggests that the maize niche was relatively stable in the region while wheat production was more marginal, and cotton had a very limited niche that required additional agricultural technologies called gravel mulch fields. These results are supported by archaeobotanical data and provide an additional line of evidence to support a slower adoption of wheat relative to maize versus a quicker adoption of sheep (wool) relative to continued cotton cultivation by Ancestral Pueblo communities in the Northern Rio Grande. The model also identifies which villages may have been better positioned to meet Spanish taxation requirements.

Keywords Arid-land agriculture · Multi-cropping · Paleoenvironmental modeling · American Southwest Pueblos · Spanish colonization

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Introduction

When Spanish expeditions arrived in the Northern Rio Grande, their chroniclers noted the agricultural productivity of the villages they interacted with, recording estimates of yields and storage, as well as commenting on differential agricultural abundance among the villages. Reports of Pueblo agricultural productivity in the early expeditions are varied, with the highest estimate of productivity (from Coronado's 1541 expedition) stating that Pueblo people had been able to store enough maize to sustain the community for seven years (Vlasich, 2005) and a lower estimate (from Gaspar Castano de Sosa's 1590–1591 journal) stating that one year's crop would last two years (Schroeder & Matson, 1965). It is important to keep in mind that when the early expeditions arrived, the New Mexico Pueblos were dealing with the multi-year droughts of the late 1530 s and multi-year severe, extended winters in the 1540 s (Schroeder, 1972). The reports are consistent in the reporting of which crops were grown (corn, beans, squash, and cotton, primarily) (Vlasich, 2005). Gaspar Castano de Sosa's journal from 1591 states that the Pueblo people at Cuyamungue shared the following agricultural products with the Spanish upon their arrival: maize, flour, beans, squash, tortillas, and turkeys, "all in great abundance" (Schroeder & Matson, 1965:115). The Spanish explorers requisitioned food and clothing from the pueblos, which caused friction, and also introduced new crops and livestock (Barrett, 2002). This paper explores how villages met increased demand and navigated environmental constraints through niche construction.

Niche-construction theory (NCT) recognizes the long-term entanglements of societies, organisms, and landscapes. NCT can be applied to Varien's (1999) and others' concept of community catchments, namely the standard radius (two-kilometer in the American Southwest, for example) around a village that contained the fields and resources that helped meet the subsistence needs of the community. Place-based agricultural potential models draw on these two concepts, by giving a distance-based starting point for the niche and providing a framework to integrate environmental affordances and agriculturalists' agency to establish a procurement niche. This article provides a coding framework and list of agronomic and cultural considerations to assist readers in identifying likely cultivation niches and developing an agricultural potential model for their region/time period of interest and utilize that model to assess historical questions. The questions we wanted to explore for the Northern Rio Grande case study included the feasibility of supporting the population just utilizing the passive niche, the productivity/needs explanation of active niche, village differentiation (agricultural specialization) that may align with historic accounts and Tewa tradition, and understanding what some villages contributed to the regional exchange system.

Niche Construction Theory

The evolutionary success of humankind stems in large part from our ability to intentionally modify the world around us in strategic ways. Among the most consequential activities are those related to food production. Plant and animal mutualisms,

agricultural practices, and especially the development of intensive, large-scale agricultural systems, have dramatically altered the planet through cumulative, persistent, and often irreversible changes. NCT's attention to these processes, including ecological, social, and historical contexts, makes it a particularly useful framework to help interpret agricultural change and archaeological agricultural feature and artifact assemblages. The NCT approach situates anthropology within the broader biosciences: it highlights the place of humans in and as part of natural systems; demonstrates the scale, scope, and importance of ecological inheritance; and transcends time, illuminating complex causal relations between past and present states (Quintus & Allen, 2023). The analysis of agricultural practices as emergent, continuously unfolding, but historically situated phenomena, as NCT holds, has particular promise for new insights into the complex causal networks of past ecological relations and land use (Quintus & Allen, 2023). While NCT traditionally and still often refers to the co-evolution of cultures and environments as part of cultures' efforts to domesticate specific plants and animals and increase the success of those domesticates (Smith, 2007, 2011; Zeder, 2016), recent work has started expanding the focus of NCT to later agricultural developments and the socioeconomic contexts of those developments (Altman & Mesoudi, 2019; Boivin *et al.*, 2016). Because NCT highlights environmental use practices, situates them in historical cultural socioeconomic context, and focuses on change and effects, it is a useful framework for agricultural potential models and understanding agricultural feature construction.

Agricultural Niche Construction

Recent developments in agricultural niches and the concept of NCT have moved beyond a focus on food to food systems, factoring the production and consumption aspect into updated niche constructions (Reed, 2021). As a result, cultivation niche models utilizing a NCT approach cannot rely on agronomic parameters alone.

To understand agricultural intensification as a form of niche construction, researchers need to be able to distinguish passive agricultural niches, where the phenological needs of particular crops were met by generally unaltered environmental conditions, from active niches, which are spaces where producers exerted additional effort to augment the landscape to enable crop growth (Aaby & Desmond, 2021). Base needs of crops and base ways that humans manipulate (*e.g.*, cultivate) plants are part of the passive niche. Active niches involve constructing an environment to meet the plant's needs in the absence of a sufficient passive niche to meet production expectations. This paper focuses on the phenological needs of water and heat. We acknowledge that other researchers may be interested in additional parameters such as soil quality. Other researchers could use the way we model active niche characteristics to include their parameters of interest.

Farmers can increase agricultural production in two ways: by utilizing more land (extensive strategies) or by investing more energy on the same land (intensive strategies). Extensive strategies tend to align with utilizing more of the passive niche, whereas intensive strategies tend to align with modifying the landscape and investing additional effort to create an active niche. Agricultural feature construction is an

example of an intensive strategy that is part of the active niche. Only by overlaying the passive and constructed niches in a cultural context and triangulating these three aspects can one make inferences about where these cultivation techniques might exist and why. The model and case study presented here provide a framework for that triangulation and an example of resulting inferences.

Agricultural niche reconstruction efforts in the North American Southwest have primarily focused on maize, which limits the ability to study food and agricultural systems with tradeoffs between different agricultural products (such as maize and wheat, or cotton and wool) during times of socioeconomic change, such as colonization. Studying and understanding agriculture is important for understanding colonial dynamics and colonization processes and effects. Indeed, agriculture is a key component of the third of the three "critical variables" identified by Lightfoot *et al.* (2013) that are critical to understanding colonial dynamics. These three variables are polity size, polity structure, and landscape management practices. In challenging climates such as the Southwest, agriculture is closely tied to landscape management.

Pueblo Agricultural Innovations

For the Tewa-speaking Pueblos along the Rio Grande in northern New Mexico and other ancestral Pueblo people, the dryland farming system they developed operated relatively uninterrupted from the AD 1200 s migration discussed in Tewa oral tradition until the arrival of the Spanish in the area in AD 1540, led by Francisco Vazques de Coronado (Barrett, 2002; Vlasich, 2005). Important Indigenous crops included *Zea mays* (maize) for food and *Gossypium* sp. (cotton) for clothing. This system sustained the Pueblos and helped facilitate trade between the Pueblos and Great Plains Tribes (Spielmann, 1991). Settlement spacing and agricultural feature distribution research have resulted in an average of a two-kilometer radius around each village for its agricultural catchment, with some villages expanding up to a four-kilometer radius (Boerger, 2025; Davis, 2022; Reese, 2021; Varien *et al.*, 2007).

The Spanish arrival brought new crops, such as *Triticum sp.* (wheat), and new domesticated animals, such as *Bos* sp. (cattle) and *Ovis* sp. (sheep), to New Mexico, in addition to new economic systems (taxation), belief systems (Catholicism), and rules of government, ownership, and control (Trigg, 2005). These rules and economic systems were formalized and enforced upon the Pueblos with the establishment of the Spanish colony of New Mexico in 1598, under the governorship of Don Juan de Oñate (Trigg, 2005). Pueblo people resisted the oppressive aspects of Spanish colonial control, pushing the Spanish out of New Mexico in the Pueblo Revolt of 1680, regaining Pueblo control of the area until the Spanish reconquest under de Vargas in 1692 (Liebmann, 2012; Spielmann *et al.*, 2009).

Numerous accounts document significant incorporation into Pueblo life of colonial agricultural practices and products, such as wheat, evident by the AD 1700s–1900s (Ford, 1972, 1992; Mills, 2008; Trigg, 2005). Evidence of wheat adoption from that period includes the use of Tewa Pueblo words for wheat and bread ovens instead of Spanish loan words, and mentions of wheat in Tewa traditional songs (Bernstein & Ortman, 2020; Bernstein *et al.*, 2019). Pueblo production

and consumption of ceramics to prepare and serve introduced crops and dishes following the Pueblo revolt also increased during this time, as evidenced by increased presence of and use-wear on these vessel types. However, how quickly and abundantly these introductions were adopted following Spanish contact, and how regimented that incorporation process was, are less well understood.

The archaeobotanical samples from four ancestral Tewa Pueblo villages dating shortly before and at the start of Spanish colonialism contain very limited evidence of Pueblo adoption of Spanish crops (Davis *et al.*, 2025). This could be due to accumulation limitations (*i.e.*, not enough time of colonial crop presence had passed for enough wheat or other Spanish crop material evidence to accumulate in these initial centuries of contact to be detectable in the archaeological record) or it could demonstrate that Pueblo communities did not adopt wheat readily in these areas and that it was adopted and experimented with more gradually in Pueblo farms over the initial decades of colonialism, not becoming fully incorporated into Pueblo farms until after the seventeenth century. The presence of the livestock signature *Sporormiella* in pollen samples as well as faunal data from Cuyamungue and San Marcos do indicate that livestock was more readily adopted than wheat, or at least that there is more evidence of it in the record from the sixteenth and seventeenth centuries in these Pueblo communities (Davis *et al.*, 2025). There are Pueblo cultural reasons that this would make sense, such as the importance of maize in Pueblo culture and ceremony and the relative efficiency of raising wool-bearing livestock compared to growing cotton. These data can be compared to crop ratios from contemporaneous Spanish settlements in New Mexico. Dawson (In Press) compares crop ratios from contemporaneous Spanish settlements in New Mexico and finds that Spanish settlements show a higher ratio of wheat to maize, with relatively limited maize adoption. Contextualizing the archaeobotanical and faunal patterns considering Pueblo cultural understandings of particular plants and animals—while also considering Spanish regulations—can highlight Pueblo agency and decision-making pertaining to agricultural practice and land use during the contact era.

Gravel Mulch Field Technology as Active Niche Construction

Recent archaeobotanical and paleoenvironmental research has demonstrated that cotton was often grown in anthropogenically modified areas known as gravel mulch fields (Camilli *et al.*, 2019; Eiselt *et al.*, 2017; Kessler, 2020). They are often located at the tops of mesas or ridges, ringed by associated features known as barrow pits. Gravel mulch fields are cleared of growth-impeding obstacles and "mulched" with well sorted small local quaternary gravels and are often demarcated by borders of river cobbles (Fig. 1). Quaternary gravels can consist of resistant quartzite, sandstone, and igneous, metamorphic, and volcanic rocks and reflect alluvial activity during the Quaternary Period (2.6 million years ago to present) (Scott & Moore, 2007). Gravel mulch fields would have functioned to regulate temperature and water in ways that might have opened up new areas for cultivation and/or extended the growing season to allow for crops such as cotton to be grown. Dominguez (2002) quantified that one needs to get 7–12 years of improved yield out

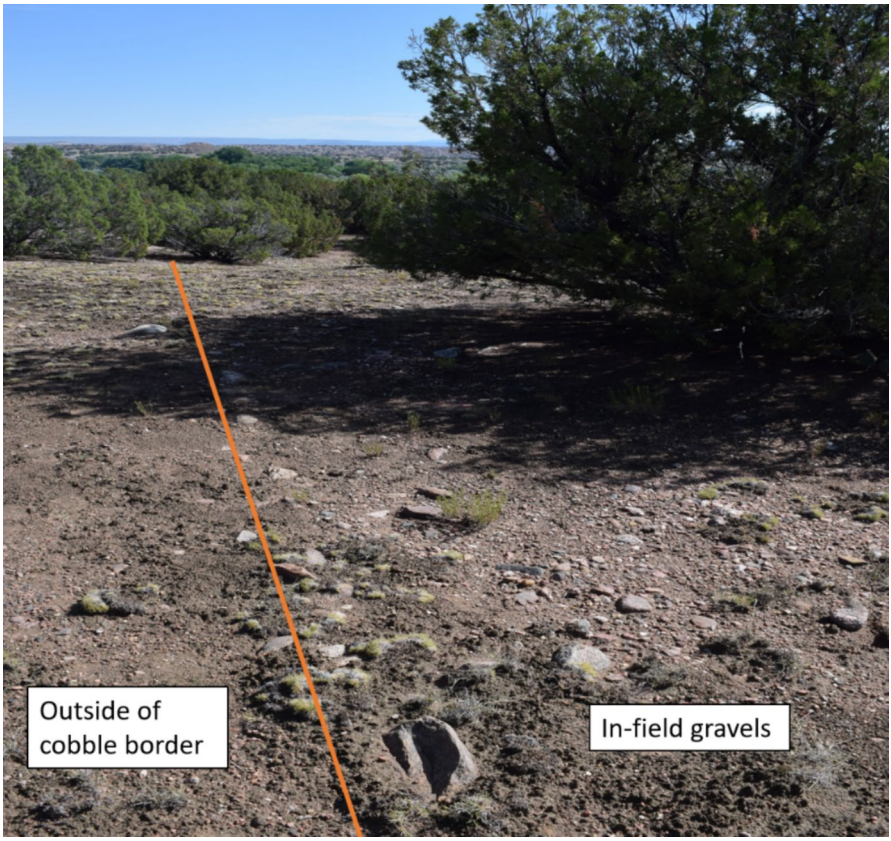

Fig. 1 Gravel mulch field with cobble border

of the mulch and bordered fields to make construction investment worth it, so, once constructed, fields were likely used for at least that duration.

Because of their abundance, visibility, preservation, continued impact on the landscape, and compatibility with other archaeological methods such as pollen and phytolith analyses, gravel mulches and their associated cobble borders have featured regularly in archaeological literature and debates (Bocinsky *et al.*, 2016; Camili *et al.*, 2019; Davis, 2022; Dean, 1995; Dominguez, 2002; Eiselt, 2019; Kessler, 2020; Lightfoot & Eddy, 1994; Lightfoot & Eddy, 1995; Moore, 1995; Periman, 1996). Scholars agree on their construction, but the conversation on their utility and which environmental constraints they were designed to overcome is evolving. Eiselt and colleagues (2017), as with Duwe *et al.* (2016), argue that gravel mulch fields may increase and/or regulate surface or subsurface temperatures and conserve water as described below. Others argue that their primary purpose was to conserve water (Periman, 1996). Lightfoot and Eddy (1994) argue that gravel mulch was applied especially during periods of drought to reduce soil erosion from wind and water, increase soil temperature to extend the growing season, increase water infiltration,

and reduce the evaporative loss of water from wind and sun. For example, they demonstrate increases in crop biomass and yield with the use of gravel mulch technology (Lightfoot & Eddy, 1994). The choice of gravel versus other organic materials is important: although organic mulch would also conserve water, it reduces temperature. Other scholars suggest gravel mulch enhances subsurface water flux and that locations were chosen to take advantage of monsoon rains (Kessler, 2020). Studies have also noted that gravel mulch fields located on northern slopes retain winter moisture (Eiselt, 2019). The gravel mulch fields in this study area (see also Davis, 2022) were found in a variety of locations, including those discussed by both Kessler and Eiselt. Davis *et al.*'s (2024) thermal imaging study of gravel mulch fields showed significant differences in the temperature signature (higher) in cobble-bordered gravel mulch fields compared to the surrounding landscape.

Cobble borders surrounding the fields also play a functional role for water retention and temperature modification (Dominguez, 2002). The slope and aspect of a hill can help maximize water input in an area as snow accumulates and prevent the loss of water. The size and spacing of cobbles can further contribute to water retention. The addition of mulch was added near the cobble borders to thicken the edges and control input during intense rainfall events. The field grids were constructed in dimensions to control a year's worth of water inputs (Dominguez, 2002). These grids provided an optimal solution to the need to maximize the input of water and yet minimize the investment required for construction (Dominguez, 2002). Other advantages include the slowing of wind, concentration of runoff, and holding and regulating moisture (Lightfoot & Eddy, 1994). Despite debates about whether temperature or water control was the predominant incentive for gravel mulch field construction, researchers agree that these were active niche features.

Irrigable Area Use as Active Niche Construction

Leading up to and including the AD 1500 s, upland villages in the Rio Grande were depopulated and people coalesced on the Rio Grande floodplain. A few examples of pre-Spanish Pueblo ditches are also known; these ditches were established in the 1400 s and 1500 s based on surrounding temporally diagnostic ceramics (Barnes, 2010; Lightfoot, 1990). The Spanish acequia system was used by Pueblo and Spanish people, constructing irrigation channels to expand the irrigable area in the Rio Grande corridor.

Pueblo Agriculture and Spanish Colonization

The agricultural potential model for maize, wheat, and cotton which is the subject of this article, is part of a larger project that examines how Pueblo agriculture and land use changed from the Late Pre-Hispanic Period through the initial period of Spanish colonization (AD 1300 s to 1680). This project, with guidance from Pueblo community members, land management agencies, and private landowners, uses surface survey, a new core sampling methodology, and paired pollen and phytolith analyses to address this topic in the agricultural catchments around four New Mexico pueblos

as part of a minimally invasive, collaborative, and community-directed archaeology effort. Analyses of changes in the location, type, size, and density of the agricultural features found on survey, as well as changes in crop growth revealed through analysis of the core samples, enable a better understanding of Pueblo agricultural adaptations over time and the extent to which Spanish plants, animals, and agricultural methods were incorporated into Pueblo agriculture. More broadly, the lessons learned from this research provide insight into sustainable climate-adapted agricultural strategies with continued usefulness for small-scale arid-lands communities.

As discussed previously, the archaeobotanical evidence (pollen and phytolith samples) obtained from soil cores in agricultural fields at the four case study sites suggested a slower uptake of wheat compared to livestock. Those samples also showed that maize was prevalent across all four agricultural catchments and all field types, whereas cotton tended to be constrained to gravel mulch fields (Davis *et al.*, 2025).

Surface survey data demonstrated differences in agricultural investment among villages, as evidenced by the number, size, and distribution of agricultural features, not necessarily correlated with population size (Davis, 2022). Temporally diagnostic ceramic sherds in fields indicated that the labor-intensive gravel mulch fields, at least at Cuyamungue, may have been disused in favor of floodplain farming or near-village fields without gravel mulch. If the gravel mulch fields were, in part, to expand the cotton growing niche, this disuse would also make sense in light of the uptake of sheep for wool.

A Pueblo Case Study from the Northern Rio Grande

The Northern Rio Grande's agricultural focus and variety of crops make it a useful case study for modeling agricultural potential and the use of niche construction to meet socioeconomic demands. The four ancestral Pueblo communities (referred to in an archaeological context as "sites") of focus in this study are shown in Fig. 2 and are from north to south: Poshuouinge, Cuyamungue, San Marcos, and Pueblo Blanco. Figure 2 also shows the locations of other Northern Rio Grande Pueblo sites and the study area included in the agricultural potential model. The distribution of these sites controls for microenvironmental and social (colonial influence) variation and is representative of the diversity of the Northern Rio Grande region in terms of extensive and intensive agricultural strategies and passive and active niches. They also represent a range of degrees of non-agricultural economic production (*e.g.*, some villages being known for exporting certain goods while others were not).

Case Study—Data and Methods

The Northern Rio Grande

The Eastern Pueblos are concentrated around the Northern Rio Grande area of New Mexico, near present-day Santa Fe and Albuquerque (Adams & Duff, 2004). The

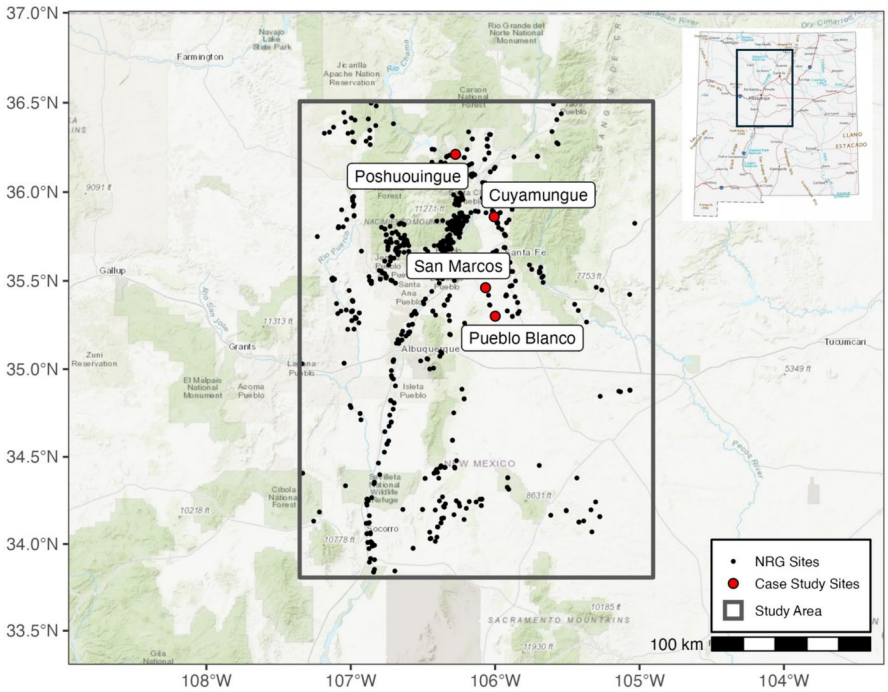

Fig. 2 Study area map

Northern Rio Grande region, or the Rio Grande Pueblo region, is where the greatest concentration of settled Indigenous farming villages existed during the period of interest (AD 1400s–late 1600 s; Barrett, 2002). The boundaries of this area are considered to extend along the Rio Grande rift valley from Taos Pueblo in the north to Senecu in the south, extending east to Pecos Pueblo and west to Jemez Pueblo, covering what is today north-central New Mexico (Fig. 2; Barrett, 2002; Ford, 1992). Others have defined the Northern Rio Grande as extending along the Rio Grande from near the present-day Colorado-New Mexico state line in the north to the confluence of the Rio Grande and Rio Puerco in the south to the Canadian River on the east and the Rio Puerco and Rio Chama on the west (Wendorf, 1954). However defined, the Northern Rio Grande had population concentrations in various basins: including the Tewa Basin, the Galisteo Basin, and the Socorro Basin, each inhabited by a particular language group. Our work is concentrated in the Tewa and Galisteo Basins.

The physical environment of the Northern Rio Grande is fairly characteristic of a semi-arid high-desert environment. The topography is a mix of mesas, canyons, and valleys bounded by mountains and mesas. Much of the rock is sedimentary sandstone. The soils are fairly thin and not ideal for dry land farming, but they are suitable for plant growth and dry land farming is still possible, if limited, in some areas (notably the Pajarito Plateau on the eastern flank of the Jemez mountains; Bocinsky & Kohler, 2014). Precipitation is around 12 to

14 inches per year on average, with the majority of rainfall occurring in July. Rainfall and temperature can be highly variable, particularly with changes in elevation. The vegetation and animals found in the study area are characteristic of the Upper Sonoran life zone, where the Great Basin, Plains Grasslands, and Juniper Parkland come together. Vegetation is a relatively sparse mix of grasses, bushes, yucca, sage, and juniper in the higher elevations and some willows and cottonwoods and thicker vegetation in the floodplains. Birds, coyotes, rabbits, and mule deer are common (Anschuetz, 1995; Kurota, 2006).

Case Study Sites

The inhabitants of the four case study sites likely spoke Tewa or Keresan (Duwe *et al.*, 2016; Kurota, 2006; Bernstein and Ortman, 2020; Welker, 1997). Within each site, the field team surveyed three microenvironments within the two-kilometer-radius agricultural catchment around the village.

Poshuouinge ("squash place") is one of the largest ancestral Tewa Pueblo sites with an entirely pre-Hispanic occupation. The village was occupied from approximately AD 1375–1500 and consisted of three large roomblocks, each of which surrounded a large plaza area (Jeancon, 1923). The village had access to ridgetops for farming, broad flat mesa tops, and a large floodplain area along the Chama River.

Tewa people inhabited Cuyamungue ("K'uuyemugeh": "stones falling down place") both before and during Spanish contact. Ceramic evidence suggests that Cuyamungue was occupied from the late AD 1100 s until the early 1700 s (Bernstein and Ortman, 2020). The village consisted of several small roomblocks, kivas, and open areas around the village to serve as plazas. It did not have a mission church. The village is near several ridgetops and mesa tops and smaller arroyo floodplains for farming, as well as Tesuque Creek. Current Tewa pueblos have an ancestral connection to Cuyamungue.

Pueblo Blanco is a precontact Pueblo site in the Galisteo Basin, where both Tewa and Keres Pueblo people lived. Temporally diagnostic ceramics date the general occupation of the site area to AD 1300–1490, with some use as late as AD 1600 (Kurota, 2006). The village had around sixteen roomblocks, five kivas, and seven plazas, as well as reservoirs (Kurota, 2006). Pueblo Blanco had the most limited access to year-round waterways of the four sites.

San Marcos is a pueblo that was occupied both before and during Spanish contact. San Marcos was occupied from approximately AD 1340–1700 and was one of the largest adobe pueblos in the Galisteo area. It was a multi-plaza village with multi-story structures, estimated at 2000 rooms in 40 roomblocks (Welker, 1997). It was known as a ceramic production center (Ramenofsky & Schleher, 2017). During the Spanish colonial period, a mission church was located in the village. The extent of its agricultural features is the largest documented in the Galisteo Basin, including gravel-mulch fields, bordered gardens, gridded gardens, borrow pits, and check dams (Lightfoot, 1990).

Modeling Crop Niches for Maize, Cotton, and Wheat

Data Overview

To assess the agricultural potential and expected passive versus active niche area and begin to make inferences about the degree and nature of contribution of the case study villages to the regional agricultural economy, the authors generated datasets for passive and active niche characteristics of the four case study villages and the Northern Rio Grande region more generally. Datasets generated include precipitation (with crop-specific threshold options), growing degree days (with crop-specific threshold options), gravel mulch potential, irrigable areas, and potential grazing areas. The PPT and GDD datasets (passive niche) were derived from PaleoCAR. The gravel mulch potential dataset (active niche) was derived from geologic maps. The irrigable areas dataset (active niche) was derived from floodplain and river and streams data. The potential grazing area dataset was modified from U.S. grasslands datasets.

Phenological Requirements

Fundamentally, plants require water, light, and heat to perform photosynthesis efficiently and flourish. Here, we model growing conditions for maize, cotton, and wheat as a function of the amount of water available to the developing plant, and the amount of heat that accumulates over the plant's development. In dryland agricultural settings plant-available water derives primarily from direct precipitation (PPT) accumulated over time and stored in snowpack or soils. Water requirements for crops can vary by species, cultivar, and developmental stage. Accumulated heat requirements are also variable across crops—while warmer temperatures generally lead to higher rates of photosynthesis, individual species and cultivars have different ideal temperature ranges for growth. The growing degree day (GDD) is a measure of accumulated heat within that ideal range, specific to a particular cultivar. Daily GDD is calculated as the average daily temperature within the growth range, defined as a lower and possibly upper limit for growth of the cultivar. Accumulated daily GDDs over a growing season show a strong correlation to the phenological stages of particular cultivars (Gilmore & Rogers, 1958).

GDD and PPT thresholds for cotton, maize, and wheat growth derive from historic growth trials of heirloom and traditional varieties of each crop. In this study, we selected thresholds for the plants to reach maturation and harvesting from growing environments within or similar to those in the Southwest (Mauney & Stewart, 1986 (cotton GDD); Wright *et al.*, 2015 (cotton GDD); organiccotton.org (cotton PPT); Bocinsky & Kohler, 2014 (maize GDD); Schachner *et al.*, 2021 (maize PPT); Ruiz *et al.*, 2019 (wheat GDD); Guedes *et al.*, 2016 (wheat GDD); Cann *et al.*, 2020 (wheat PPT)). The thresholds used are presented in Table 1. Outside of the Southwest, maize may need 30 cm of precipitation to reach maturity, but growth trials with heirloom varieties cultivated by Pueblo people in the Southwest are adapted to less precipitation (Fig. 3; Bocinsky & Kohler, 2017; Schachner *et al.*, 2021). Determining the areas in New Mexico where Ancestral Pueblo people were growing Gossypium hirsutum

Table 1 GDD and PPT thresholds

Crop	GDD Niche Threshold (C GDD)	GDD Base Temp (degrees C)	GDD Cap Temp (degrees C)	PPT Niche Threshold (cm)
Cotton	1690	15	N/A	14
Maize	1000	10	30	25
Wheat	1000	0	25	30

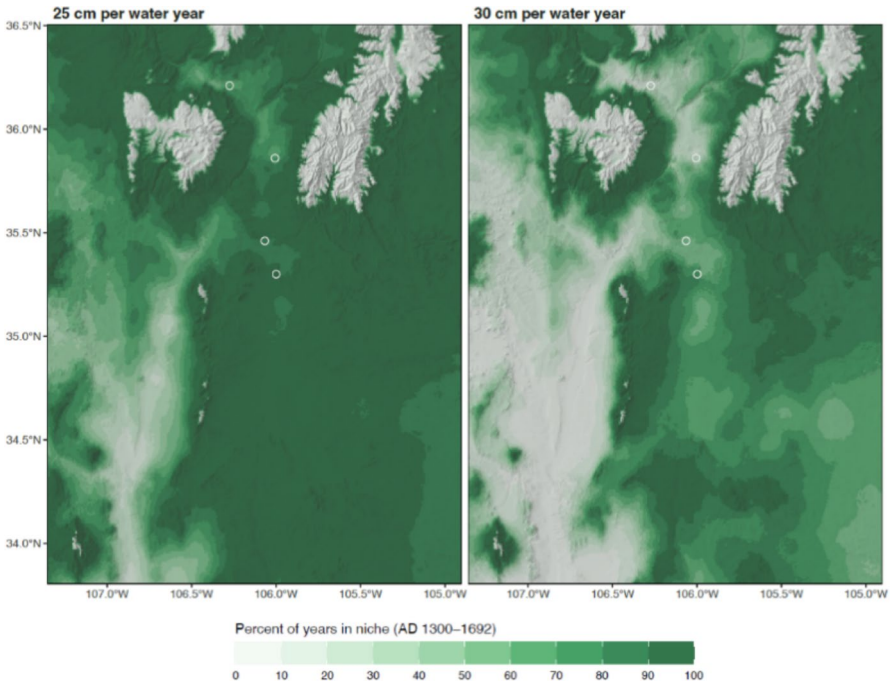

Fig. 3 Percent of years in maize niche based on 25 cm versus 30 cm of PPT

versus Gossypium hopi is still an open area of research (Lewton, 1912; White, 1941). For this study, Gossypium hirsutum agronomic data were used due to a greater availability of and number of comparative experiments for those data.

PaleoCAR Reconstruction of Passive Crop Niches

The GDD and PPT data used in the model derive from paleoclimate reconstructions developed using the PaleoCAR method (Bocinsky & Kohler, 2014; Bocinsky *et al.*, 2016). PaleoCAR uses dendrochronology data to reconstruct temperature and precipitation estimates, by year, for the Southwest since about AD 800 (Bocinsky & Kohler, 2014). Previous applications of PaleoCAR include assessing potential

maize growing productivity and assessing climate-driven risks to food security (Strawhacker *et al.*, 2020).

PaleoCAR is considered an expert model, meaning that it automates the analyses that experts in the field of climatology and dendrochronology data would do. The model selects regional tree ring chronologies that are highly correlated to a climatological variable of interest (such as annual precipitation) and then uses a linear model to estimate the relationship between the climate variable and tree growth (Bocinsky & Kohler, 2014, 2016; Bocinsky *et al.*, 2016; Bocinsky 2021). The PaleoCAR reconstructions used here are based on climate variables developed from the 800 m-resolution PRISM interpolated meteorology dataset (Daly *et al.*, 2008). Put simply, we produce annual estimates of the GDD and PPT values for each 800-m PRISM pixel within the study area each year for each of the three crops. We developed PaleoCAR-based reconstructions for water-year precipitation, and growing season accumulated GDDs for each of the crops in our study based on the thresholds above. The resulting data are annual rasters (at the 800 m PRISM resolution) indicating whether (or not) a crop met the thresholds described in Table 1—effectively binary indicators of success or failure of a harvest for a particular time and place. Our analyses consist of calculating probabilities of success or failure across space (by pixel, or within agricultural catchments) and through time (Precontact, Contact, Colonial, Post-Revolt). This broadly defines the passive niche for maize, wheat, and cotton across the region. We then explore how these passive niches could have been (and, we argue, were likely) enhanced by active niche construction strategies such as irrigation and gravel mulching.

Gravel Mulch Potential

The gravel mulch potential layer was developed using USGS geologic layer maps (mrdata.usgs.gov), selecting for quaternary gravel deposits ("Qa" and "Qp" geologic codes) in the environments where gravel mulch fields have been recorded in previous studies (ridge and mesa tops with slopes of less than 10 degrees). The gravel mulch potential layer was compared against gravel mulch field data generated from New Mexico state geologic maps (Camilli *et al.*, 2019; geoinfo.nmt.edu) and from visual observations of and physical relocations and mapping of gravel mulch fields (Eiselt, 2019) to ensure that it was a reasonable approximation of where Pueblo farmers could create gravel mulch fields. The USGS basemap was expanded to include these areas.

Irrigable Areas

The irrigable area layer was developed from the New Mexico rivers layer available through ESRI and refined using federal and state floodplain data (FEMA, 2021). We buffered the New Mexico rivers layer based on floodplain extent averages to approximate the area over which irrigation ditches likely could have brought water from the rivers to the fields. We compared this to aerial imagery and historic maps of acequias to "ground truth" the irrigable area layer.

Grasslands

The grasslands layer was downloaded from the United States LANDFIRE database, a joint initiative of the US Forest Service and the Department of the Interior (Landfire, 2022). While this layer does not factor further in the agricultural potential model, it served to inform where the areas of higher livestock grazing potential were. The introduction of livestock into the agricultural economy of the Pueblos could have influenced Pueblo land use and farming decisions.

Regional and Local Applications

For this project, the model focuses on two spatial levels: the regional level and the village level. The regional level encompasses the model area described in the previous paragraph, the extent of which can be seen in Fig. 4. This map shows the percentage of years that each 800-m pixel within the study area met the precipitation and growing degree thresholds for each of the three focal crops of the study: maize, wheat, and cotton.

The regional-level model (Figs. 4 and 5) shows growing degree day (GDD), precipitation (PPT), irrigable area, and gravel mulch area data for growing maize, wheat, and cotton, at an 800-m spatial resolution and yearly temporal resolution, from AD 1300 (the time by which all four of the focal villages were inhabited) to AD 1692 (the reconquest by the Spanish). The village-level model, presented in a later section, shows these same data for two-kilometer agricultural catchments around the four Pueblo villages that are the focus of the study. The village-level model also includes calculations for proportions of grasslands within the two-kilometer catchment and the proportion of overlap between grasslands and the farming

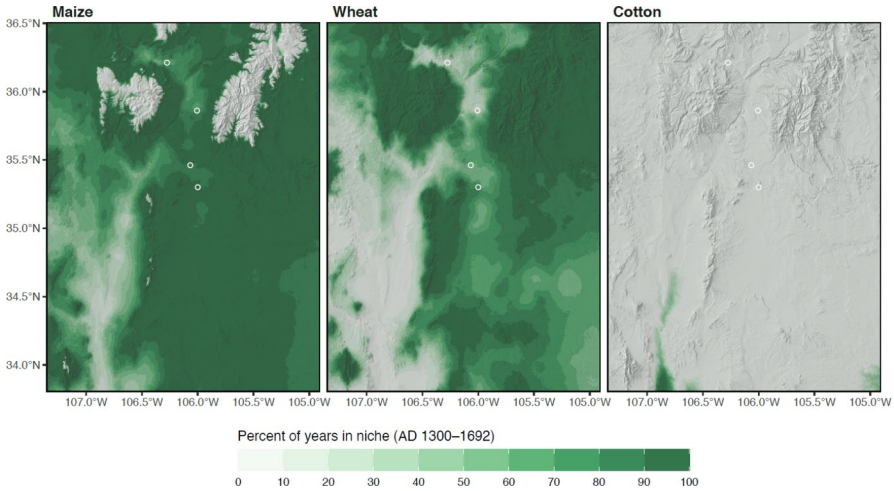

Fig. 4 Percent of years by crop in the passive agricultural niche in the NRG study area, based on GDD and PPT thresholds

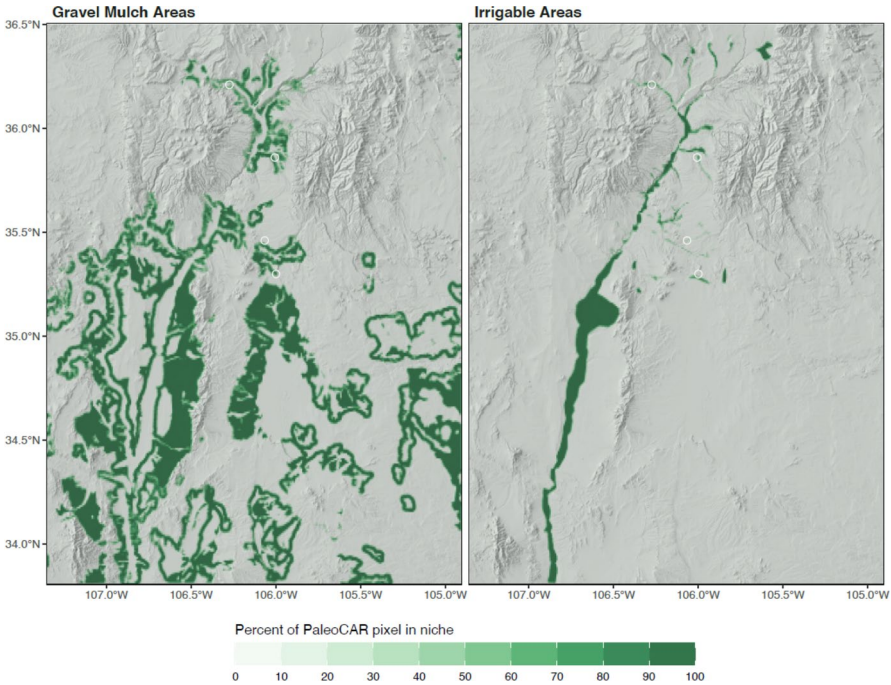

Fig. 5 Gravel mulch and irrigable areas

niches. These grasslands would be appropriate forage for grazing animals introduced by the Spanish and thus represent potential grazing area.

The gravel mulch (GVM) areas and irrigable (IRRIG) areas generated by this model are shown in Fig. 5. The fraction of a pixel that is irrigable, or has quaternary gravel, is the fraction of that pixel in the niche, and this fraction was modeled as constant over time, irrespective of annual weather. Thus, an area that has quaternary gravels or is within the calculated floodplain was considered to be within the gravel mulch niche or irrigable area niche, respectively, 100 percent of the time.

Estimating Historic Yields

The yields (see Table 2) for maize were derived from historic yield records from early twentieth-century New Mexico, from small-scale, non-industrialized farms (NRG Dataset, 2017). The yields for cotton were extrapolated from heirloom variety

Table 2 Yield estimates

Crop	Yield	Yield unit
Cotton	0.0576	lbs lint/meter^2
Maize	0.126	kg/meter^2
Wheat	0.00331	bushels/meter^2

Table 3 Proportions of catchment in maize niches

Village	Proportion in PPT Niche	Proportion in GDD Niche	Proportion in Irrigable Niche	Overall Proportion in Maize Niche (in GDD Niche and in PPT or Irrigable Niche)
Cuyamungue	0.721	1	0.353	0.850
Poshuouinge	0.673	1	0.2778	0.806
San Marcos	0.886	1	0.111	0.886
Pueblo Blanco	0.958	1	0.0526	0.958

Table 4 Proportions of catchment in wheat niches

Village	Proportion in PPT Niche	Proportion in GDD Niche	Proportion in Irrigable Niche	Overall Proportion in Wheat Niche (in GDD Niche and in PPT or Irrigable Niche)
Cuyamungue	0.239	1	0.353	0.512
Poshuouinge	0.225	1	0.2778	0.450
San Marcos	0.562	1	0.111	0.611
Pueblo Blanco	0.753	1	0.0526	0.762

yield data (bolls and plants) (Toler et al., 2019) and adjusted for the spacing and cotton weight per boll of Pueblo cotton plants according to ethnohistoric and ethnographic accounts (Fulton, 1938; Huckell, 1993). The yields for wheat came from historic yield records from nineteenth-century New Mexico (Bennett, 1938). Table 2's yield estimates assume the GDD and PPT thresholds from Table 1.

Case Study—Results and Discussion

Crop Niches for Maize, Cotton, and Wheat

The agricultural potential model uses the PPT, GDD, IRRIG, and GVM data and crop thresholds just discussed to determine the percentage of years that an 800 m pixel had the right set of conditions to grow a particular crop. To determine which conditions (PPT, GDD, IRRIG, or GVM) needed to be included in the calculations for each crop, we relied on previous agronomic and archaeological studies and the model testing program. These studies suggested that GVM was necessary for growing cotton and IRRIG was helpful in increasing the area for growing maize and wheat. The model testing program involved running the model for each crop one factor at a time and comparing the resulting niche to the areas where that crop has been known to grow based on microbotanical evidence from Davis *et al*. (In Press) and other recent studies (Kessler, 2020). The results of these tests are shown in Tables 3, 4, and 5. IRRIG was a negligible variable for cotton, since cotton was not commonly grown in the floodplain (irrigable areas), and GVM was a negligible variable for wheat and maize and thus is not included in the tables. Interestingly,

Table 5 Proportions of catchment in cotton niches

Village	Proportion in PPT Niche	Proportion in GDD Niche	Proportion in Gravel Mulch Niche	Overall Proportion in Cotton Niche (in PPT Niche and Gravel Mulch Niche)
Cuyamungue	0.996	0	0.235	0.234
Poshuouinge	0.992	0	0.389	0.389
San Marcos	1	0	0.111	0.111
Pueblo Blanco	1	0	0.421	0.421

these tests also suggest which environmental constraints GVM technology was used to overcome. For cotton, none of the study localities met the GDD niche for cotton but were well within the PPT niche for cotton. Thus, for cotton, GVM likely helped to overcome GDD deficiencies.

While the model incorporates many of the key factors Pueblo people took into account when establishing farming areas and making planting decisions, and the predominant factors that would have impacted crop yields, there are nevertheless some factors that could not be incorporated into the model. These additional factors include: people's (individual families'/groups') different responses to the environmental conditions presented in the model; interactions with neighbors or other groups that might limit which fields a family could use; pests; shifting crops and planting times (*e.g.*, crop rotations and fallow fields); post-harvest crop loss; and perennial stream access (since perennial streams were too small and irregular to be included in the irrigable area data this model uses). Another limitation of the model is that the precipitation data that it uses (derived from the tree ring data in PaleoCAR and the weather station data in PRISM) is annualized and thus cannot capture seasonality. The other limitation of the model is its 800-m resolution, which is excellent for a regional-scale analysis, but coarse for examining the individual two-kilometer-radius study localities. Because the resolution cannot be improved, as it is the best resolution currently available in the PRISM program, the reader should consider the results presented in the remainder of this chapter as representative of an overall assessment of the suitability of a site catchment, rather than highlighting specific areas within the catchment that were more or less suitable.

The GDD and PPT niches for the agricultural catchments around the four villages are shown in Figs. 6, 7, 8, 9, and 10. These figures (Figs. 6, 7, 8, and 9), along with Fig. 10, help illustrate why GVM and IRRIG were added as factors in the analyses. As shown in Figs. 6, 7, 8, and 9, none of the study localities would be within the cotton growing niche without the addition of gravel mulch (Fig. 10).

The overall maize and wheat niches are modeled as the sum of 1) the overlapping GDD and PPT niches and 2) the overlapping GDD and IRRIG niches for each crop. The overall cotton niche is modeled as the intersection of the GVM and PPT niches. IRRIG made a negligible difference in the area of the cotton niche and none of the areas met the cotton GDD threshold without the aid of GVM thermoregulation, which is assumed to have been adequate based on microbotanical data which show a close association of cotton pollen with gravel mulches (Fig. 10).

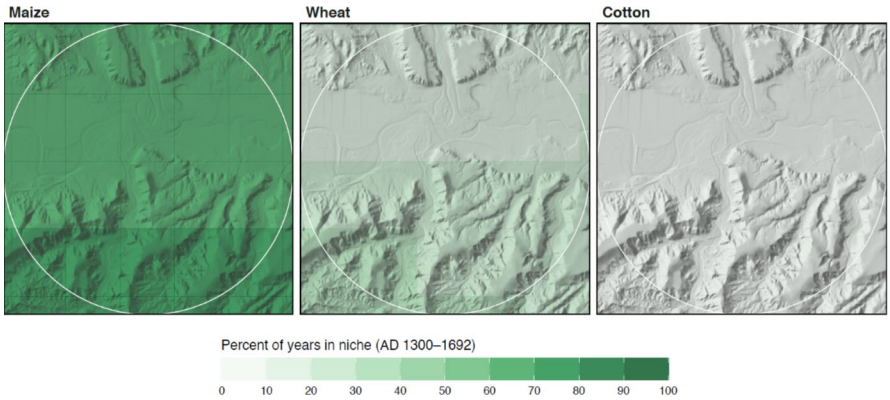

Fig. 6 Agricultural niches at Poshuouinge

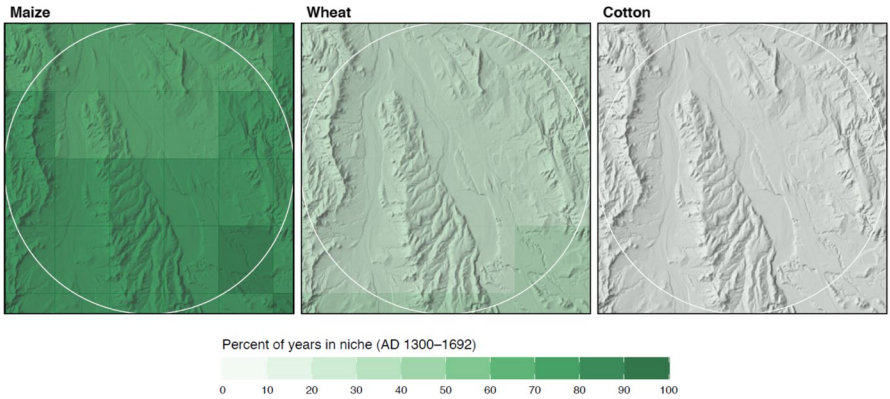

Fig. 7 Agricultural niches at Cuyamungue

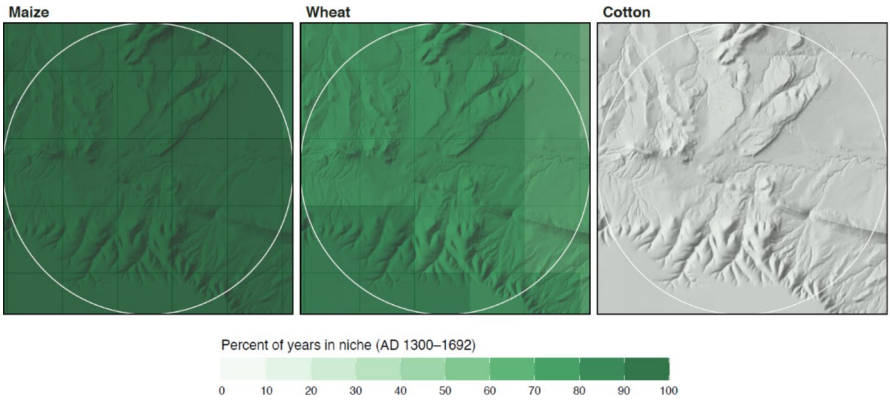

Fig. 8 Agricultural niches at Pueblo Blanco

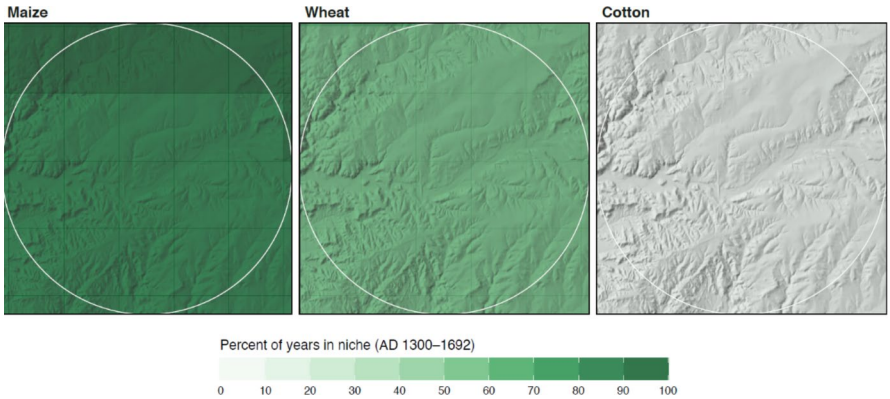

Fig. 9 Agricultural niches at San Marcos

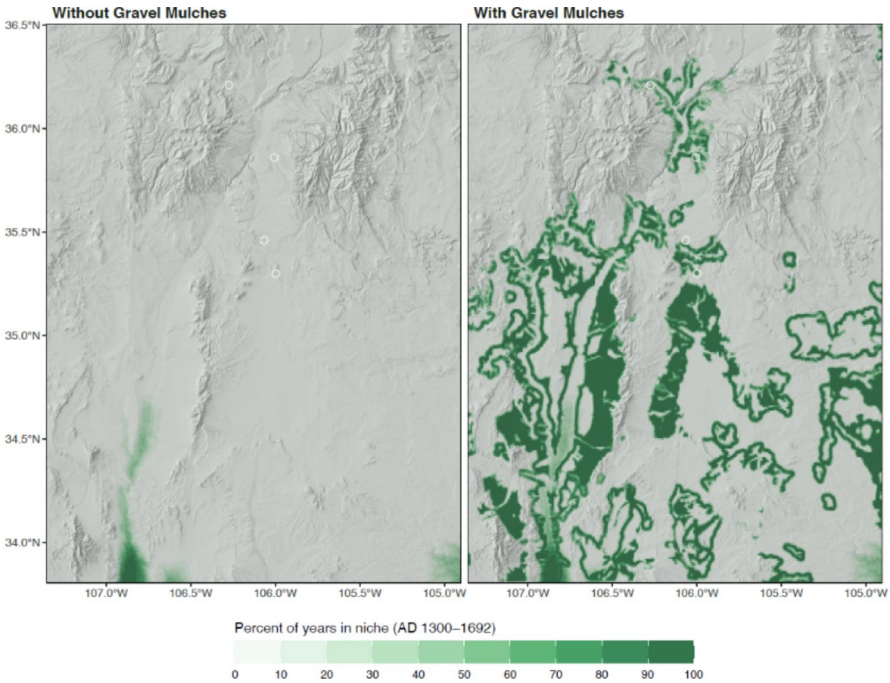

Fig. 10 Cotton niche without versus with gravel mulches

Crop Niches

The main focus of the model was to determine the proportion of years between AD 1300 and 1692 that maize, cotton, or wheat could be grown in each pixel (800 m area), and the proportion of pixels within the two-kilometer agricultural catchment around each village where this could occur. The code for the

model, available on GitHub (https://github.com/bocinsky/paleocar), was written in such a way that it can be easily edited for other crops or other years within the study area, from approximately AD 1000 to present. The code for identifying the crop niches consists of three main functions—proportion of catchment within the niche, variance, and area of catchment within the niche – which are written in such a way that agronomic variables or periods of interest for other researchers can easily be inserted. Examples of the three main functions (proportion, variance, area) are shown in Supplemental Materials. The advantage of using functions in coding is that it allows one to use the same code for different tests because it abstracts out the variables. For example, threshold_1 could be the GDD, PPT, GVM, or IRRIG threshold for any crop.

We derived the following datasets:

1. Proportion of agricultural catchment around each village within each crop niche, averaged over the whole time span of the study (1300–1692)
2. Proportion of agricultural catchment around each village within multiple (overlapping) crop niches, averaged over the whole time span of the study (1300–1692)
3. Proportion of agricultural catchment around each village within each crop niche, averaged within each time period of interest (Precontact, Contact, Colonial, Post-Revolt)
4. Proportion of agricultural catchment around each village within multiple (overlapping) crop niches, averaged within each time period of interest (Precontact, Contact, Colonial, Post-Revolt)
5. Proportion of agricultural catchment around each village within each crop niche, over the time period of occupation of each village
6. Proportion of agricultural catchment around each precontact village in the Northern Rio Grande within each crop niche, averaged over the whole time span of the study (AD 1300–1692)
7. Proportion of agricultural catchment around each precontact village in the Northern Rio Grande within multiple (overlapping) crop niches, averaged over the whole time span of the study (AD 1300–1692)
8. Proportion of agricultural catchment around each contact village in the Northern Rio Grande within each crop niche, averaged over the whole time span of the study (AD 1300–1692)
9. Proportion of agricultural catchment around each contact village in the Northern Rio Grande within multiple (overlapping) crop niches, averaged over the whole time span of the study (AD 1300–1692)

We discuss the results for each of these datasets below.

According to the outputs from the agricultural potential model, a greater proportion of the agricultural catchments around Poshuouinge, Pueblo Blanco, Cuyamungue, and San Marcos was able to grow maize because those areas fell within the maize growing niche (meeting GDD and PPT or GDD and IRRIG thresholds) over the greatest number of years. For each of the four case study villages, the environmental conditions for growing wheat were met in a smaller fraction of the

Table 6 Proportion of area in individual crop niches across modeled years (AD 1300–1692)

Study locality	Maize (GDD, PPT, IRRIG)	Wheat (GDD, PPT, IRRIG)	Cotton (GVM, PPT)
Poshuouinge	0.806	0.450	0.386
Cuyamungue	0.850	0.512	0.234
Pueblo Blanco	0.953	0.762	0.421
San Marcos	0.881	0.611	0.111

Table 7 Variance of proportion in crop niches across modeled years (AD 1300–1692)

Study locality	Maize	Wheat	Cotton
Poshuouinge	0.0854	0.00753	0.248
Cuyamungue	0.120	0.0124	0.190
Pueblo Blanco	0.0502	0.0351	0.257
San Marcos	0.0834	0.0328	0.105

area around the village than maize (meeting GDD and PPT or GDD and IRRIG thresholds). The most restrictive growing niche around each village was cotton (meeting GVM and PPT or GVM and IRRIG thresholds) as, without the gravel mulch fields, the GDD threshold could not be met. Table 6 shows the proportions of village catchments that were within individual crop niches, and Table 7 shows the variance in proportion of the catchment in the niche across years. The latter table shows that wheat had the least variance thus suggesting that it was the most consistent of the three crops to grow in those areas where it was feasible. However, the larger proportion of area in the maize niche made maize the more reliable crop to grow overall.

This pattern held true across time periods. There were no significant differences in the proportion within the niche when separating the data by time period. In fact, the differences were very slight and almost negligible.

This pattern also held true for the most part in other study localities across the Northern Rio Grande region. Some villages' surrounding environments (two-kilometer catchments) were more conducive to growing wheat than maize, but such variation does not appear to distinguish villages that continued to be occupied into the contact era from those that were depopulated. Clearly, factors other than local agricultural catchments were responsible for the pattern of village persistence versus abandonment during the contact period.

Overlapping niches were greatest for wheat and maize, which was expected given the similar growing conditions needed for the two plants. The overlap in the maize and wheat niches was greater in the Galisteo Basin than it was in the Tewa and Chama areas. Table 8 shows the proportion of area averaged over the years AD 1300–1692 that fell into the overlapping niches around each village. The proportion of a village's catchment in an overlapping niche was calculated by multiplying the binary rasters of individual crop niches together. For example, if an area was within the maize niche it had a value of 1 on the raster. If that area was within the wheat

Table 8 Proportion in overlapping crop niches

Study locality	Maize and cotton	Wheat and cotton	Maize and wheat	Maize, cotton, and wheat
Poshuouinge	0.262	0.0879	0.225	0.0879
Cuyamungue	0.171	0.0579	0.239	0.0580
Pueblo Blanco	0.406	0.319	0.752	0.318
San Marcos	0.0966	0.0591	0.562	0.0591

niche as well (value 1), then the overlapping niche value for that area would also be 1 ($1 \times 1 = 1$), whereas if the areas were not within the wheat niche (value 0), then the overlapping niche value for that area would be 0 ($1 \times 0 = 0$).

Adding the irrigable area to the niche made more of a difference for maize and wheat than it did for cotton in terms of expanding the area where these crops could be grown, as the gravel mulch and irrigable areas hardly ever overlap. Adding irrigable area to the model expanded the growing niche more in the Tewa and Chama area than it did in the Galisteo area.

Potential Crop Yields

To estimate the crop yields from the proportions of area that fell within the crop niches, we took the area in the niche and then multiplied by the yield information listed in Table 9 (maize example shown in Supplemental Materials). Based on interviews with Pueblo knowledge keepers and farmers, the model assumes that ancestral Pueblo farmers were planting at the correct time to maximize yields, based on cumulative, localized environmental knowledge. The niches for the area function were defined as GVM and PPT for cotton, and GDD, PPT, and IRRIG for maize and wheat. 800-m resolution is coarse data to use for yield per hectare estimates, but, as previously noted, that is the current best standard of data available. Thus, the study localities' yields discussed in this section should be considered relative to each other, rather than focusing on the absolute numbers.

Table 9 Potential annual yields of in-niche areas and taxation rate based on population (bold values represent sufficient yield)

Village	Peak Pop.	Cotton Yield at Highest Pop Pd (lbs lint)	Maize Yield at Highest Pop Pd (kg)	Wheat Yield at Highest Pop Pd (bushels)	Yearly Maize Tax on Village (kg)	Yearly Cotton Tax on Village (lbs)
Cuyamungue	1,200	**168,304**	**1,345,176**	**22,357**	**25,050**	**450**
Poshuouinge	2,176	**276,320**	**1,275,543**	**20,347**	**45,424**	**816**
San Marcos	6,000	**80,384**	1,394,235	24,994	1,25,250	**2,250**
Pueblo Blanco	4,450	**305,208**	**1,508,180**	**32,656**	**92,894**	**1,672**

The incentive for increasing yields through active niche construction can be explained in the context of demand (consumption requirements). We compared these yields against ethnohistoric estimates of how much a family needed to feed themselves for a year, archaeological estimates of the populations of the communities during this time, and the amount that the Spanish were taking in taxes. The archaeological estimates of the peak populations of the communities in the Northern Rio Grande during this time were derived from room counts, house mound volume estimates, and amounts of used ceramics (cyberSW, 2022). These estimates for the four communities are shown in Table 9 and are archived in the Northern Rio Grande Database (NRG, 2017) and cyberSW (cyberSW, 2022). The estimates of yearly taxation on the community were derived from the amount a household was taxed (Barrett, 2002) multiplied by the number of households in the village. Under the encomienda tax system, each Pueblo household was taxed each year in one fanega of maize (2.6 bushels) and a piece of cotton cloth (about 5.5 square feet), or a buffalo hide or deer skin could be substituted for the cotton cloth (Barrett, 2002). In addition to taxation, field or post-harvest losses also likely further reduced the yields that Pueblo communities could retain for their own use. In maize-growing communities in Tanzania, for example, these losses can range from 20 to 30 percent (Abass *et al.*, 2014; J. Terry McCabe, personal communication March 18, 2022). One way that Pueblo farmers have tried to reduce post-harvest loss is developing additional varieties of maize focused on more precise climatic zones and resilient to common Southwest climatic issues, such as drought and cold-air drainages (Bocinsky & Kohler, 2017). Since studies have not yet quantified post-harvest loss in Southwest Pueblo farming contexts, they were not factored into the yield calculations for this study, but future studies would benefit from collecting and incorporating this data. The estimates of a family's consumption of maize, wheat, and cotton (as cloth) in a year are calculated below.

In a maize-focused diet model, 70 percent of an Ancestral Pueblo person's diet would be maize-based (Ermigiotti *et al.*, 2020). However, microbotanical and faunal evidence suggest that other food items, such as amaranth, game, and livestock, may have factored more heavily into the Ancestral Pueblo diet than the maize-focused model states, especially during the contact time period of this study (Frank, 2002; Turner *et al.*, 2021). For a 2,000 cal per day diet, a person should consume 25 g of fiber to maintain healthy food composition ratios (Miketinas *et al.*, 2019). Maize is a high-fiber food, containing about 3.5 g of fiber per ear. If the 70 percent maize diet model is reduced to 50 percent to take into account Frank (2002), Turner *et al.* (2021), and others' findings, then 12.5 g, approximating four ears of maize, should be consumed per person per day.

If a person consumes four ears of maize per day, that is the equivalent of two pounds of maize per person per day, since an eight-inch ear of maize contains approximately 0.5 pounds of grain (Berger, 2017). Scaling this up to a year, a person consumes 730 pounds of maize, or 331 kg. Multiplying those kilograms (kg) by the village population results in 397,200 kg of maize being consumed yearly at Cuyamungue; 722,432 kg at Poshuouinge; 1,992,000 kg at San Marcos, and 1,477,400 kg at Pueblo Blanco. Comparing that to the estimated yield from the proportion of area in the agricultural niche shown in Table 9, Pueblo Blanco would have been unable

to support its community's maize needs from its own two-kilometer catchment, only producing approximately 50 percent of its community's need. San Marcos would have been close to, but not quite able to meet the four ears a day requirement with its maize yields, but would have been able to meet community needs if the requirement were three ears a day and maize did not comprise as much of the diet as has been previously thought. However, the addition of Spanish taxation on maize would have stressed San Marcos's maize supply more than Cuyamungue's due to Cuyamungue's projected yields exceeding the community's consumption needs.

If a person consumes at least 300 pounds of grain (wheat) in a year (Mentzer Morrison et al., 2010) and there are 56 pounds in a bushel, that means that a person consumes 5.36 bushels of grain (wheat) per year. Multiplying those bushels by the village population results in 6,432 bushels of wheat consumed at Cuyamungue; 11,663 at Poshuouinge; 32,160 at San Marcos; and 23,852 at Pueblo Blanco. All villages except San Marcos could have met their wheat consumption needs. Note that these maize and wheat calculations are looking at each individually; once the communities were consuming both maize and wheat, the amount of each consumed would be lower than if the other was not present. The yield for each would also be lower, since some amount of the niche would be used to grow the other crop.

Cotton manufacturers estimate that a cotton skirt and shirt require 1.5 pounds of cotton lint to produce and a cotton shirt and pants require about 2.0 pounds of cotton lint to produce (California Cotton Ginners and Growers Association, 2021). If a woman has four new outfits each year that would mean that she used 6 pounds of cotton lint each year. If a man has four new outfits each year that would mean that he used 8 pounds of cotton lint each year. Assuming that half the village population were men and half were women, that means that Cuyamungue would have used 8,400 lb of cotton per year to clothe its community; Poshuouinge would have used 15,232 lb; San Marcos would have used 42,000 lb; and Pueblo Blanco would have used 31,150 lb. Under this scenario, all the villages would have been able to support their cotton needs in-house even with taxation, unless the estimate of the amount of clothing made for each person each year is lower than the reality was.

General observations from these data discussed above and presented in Table 9 include that Cuyamungue had the highest ratio of yield to population. Poshuouinge and Pueblo Blanco have higher potential cotton yields than Cuyamungue and San Marcos do, which correspond to the greater fraction of catchment in the cotton niche at those two study localities, as shown in Table 9. The margin of surplus for maize is greater than for wheat. This may help to explain the slower and more limited shift from maize to wheat over time, in addition to the cultural importance of maize. The communities were able to support themselves for the most part even with the addition of taxation. Specifically, the addition of taxation did not change whether or not a study locality had a high enough crop yield to sustain its population, as shown in Table 9 (bold represents sufficient yield and non-bold represents insufficient yield). However, as Trigg (2005) notes, if the Spanish abused the taxation system and took taxes more than once a year instead of yearly or took more than the amount indicated in the colonial documents, the communities would have been further stressed. Differential agricultural effort and surplus in the different villages could reflect different contributions to the regional exchange system. For example, San Marcos, a

known ceramic production and export village, could have their agricultural needs supplemented by Cuyamungue in exchange for ceramic vessels, which would align with Tewa understandings of and archaeological evidence of exchange between these two sub-regions and align with Spanish historic accounts of the agricultural productivity of certain villages standing out.

Comparison of Modeled Agricultural Niches to Pueblo Fields

Extracting the proportion of agricultural features by whether or not they were within the environmentally identified niches showed that, with the exception of Pueblo Blanco, at least 63% of the farming features were within the environmentally identified niches (Table 10). This shows that the model does a fairly good job at identifying where communities were actually planting, which supports how the model was constructed and the parameters used. The agricultural features that were outside of the environmentally identified niches could have been outside of the niche because the communities were using technologies that overcame environmental constraints and thus expanded the possible growing area, or because there were other socioeconomic reasons for planting in these locations (family land claims, planting in multiple environments to reduce risk of crop failure, *etc*.). Extracting the surveyed area by the environmentally identified niche showed that at least 78% of areas surveyed to be were within areas of potential production for at least one crop in all communities. This shows that the survey did encompass probable farming areas. The percent of potential area in production that was actually in production (within the surveyed area) was calculated by multiplying the proportion of agricultural features within the niche by the proportion of survey within the niche. These data follow the population rank of the villages except for Pueblo Blanco, which as we have seen is an outlier by a number of other measures. In other words, a village with a greater population had more agricultural area in production, as would be expected. The placement of particular fields and agricultural features within the agricultural niches can be observed in the interactive map.

Grasslands, or potential grazing areas, did not frequently overlap with agricultural (potential farming) areas (Table 11). This could help explain the advantage of adding livestock to the Pueblo agricultural system because the livestock were not taking away from crop land and they could eat down the remnant stalks and grains in the fields (for maize and wheat) after the crops have been harvested. Another advantage

Table 10 Potential versus actual planting area by study locality

Study locality	Total area of surveyed features (m^2)	Proportion of features in ag niche	Proportion of surveyed area in ag niche	Percent of potential production area actually in production
Cuyamungue	8048.83	0.732	0.995	73.55
Poshuouinge	4513.16	0.631	0.775	81.41
Pueblo Blanco	9101.8	0.396	0.962	41.15
San Marcos	4120.28	0.792	0.864	91.63

Table 11 Potential grazing area by study locality

Study locality	Proportion of site catchment in grassland/grazing area (although we now know fields were grazing area)	Proportion of field feature and grassland overlap
Cuyamungue	0.136	0.052
Poshuouinge	0.176	0.439
Pueblo Blanco	0.125	0.027
San Marcos	0.019	0

of livestock is that they require less active effort from farmers to grow than does a crop that requires intensive technologies, such as cotton in gravel mulch fields. This limited grazing area close to the village (with the exception of grain remnants) mirrors patterns in other parts of the world, such as Mesopotamia, where farming generally occurred closer to the village and grazing occurred further away, beyond the farming area (Altaweel & Palmisano, 2019). Thus, livestock could be argued to be part of the passive niche.

Regional Comparisons

The same crop niche calculations performed on the four study localities to assess the proportions of the two-kilometer catchments in the maize, wheat, and cotton niches were also performed for sites across the Northern Rio Grande. Figures 11 and 12 show the distribution of proportions in the crop niches for the Northern Rio Grande region. These distributions demonstrate that the patterns of relative suitability in the four study localities, with the proportion in the maize niche being the greatest, followed by wheat, then cotton, holds true across the region. Indeed, the study locality

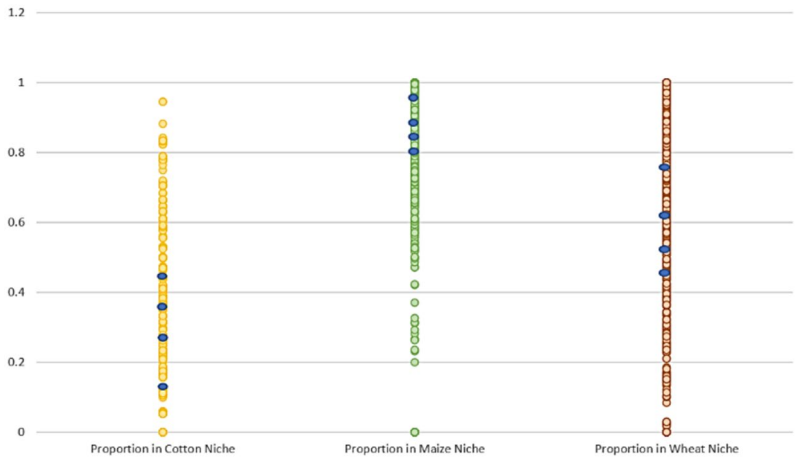

Fig. 11 NRG regional data: proportions in crop niches (Dot Plot)

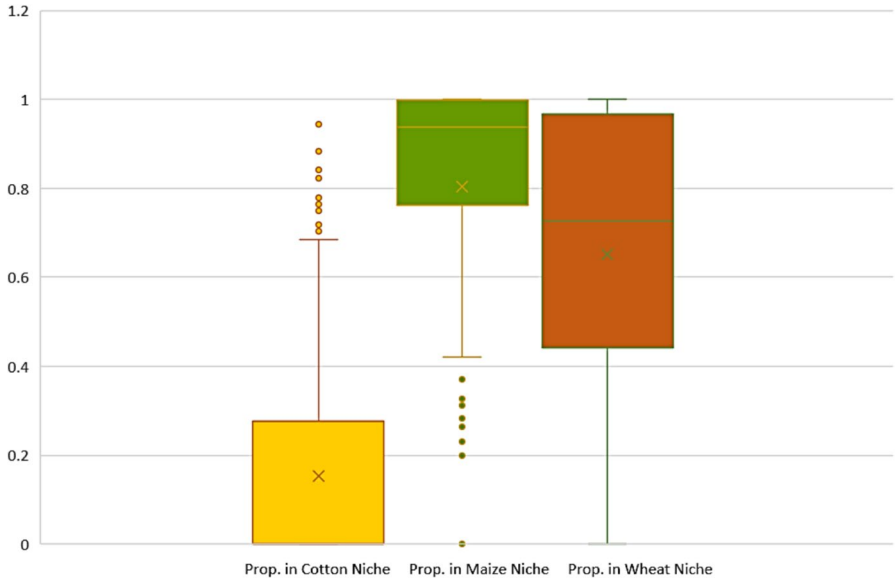

Fig. 12 NRG regional data: proportions in crop niches (Box & Whisker Plot)

data points, shown in blue in Fig. 11, show that the four study sites are representative of the regional agricultural environment. These data suggest that the environment was not the sole factor in which sites persisted into the contact era and which did not, as there was no correlation between the time period of site occupation and the proportion in the agricultural niche.

Discussion

Different crops may be more dependent on the active versus passive niches and different socioeconomic needs and social changes may influence the degree to which niche construction was required. The model shows that, in the case study region, maize could grow in the passive niche fairly well, while cotton required an active niche to grow, and wheat fell in between. The agricultural potential model applied to the case study demonstrates environmental and technological factors both impacted the ability of Pueblo people to grow different crops and helps explain where active (intensive) versus passive niches were utilized. The ability to grow maize and wheat was impacted by growing degree days, precipitation, and the amount of irrigable land available. The ability to grow cotton was impacted by precipitation and the amount of gravel mulch area available. Access to irrigable land and associated irrigation technology was more important for growing maize and wheat than it was for cotton. For cotton, the limiting factor is heat, since the GDD threshold was not met for any of the four villages' agricultural areas. This finding suggests that gravel mulch is important not just for water retention, as previous studies have reported, but also, and perhaps even more so, for thermal regulation. Other previous studies

on Southwest maize have also demonstrated the role of gravel mulch technology for mitigating GDD constraints (Bellorado, 2007, 2009). Access to irrigable land also opened up more areas for farming around Cuyamungue and Poshuouinge than it did around San Marcos and Pueblo Blanco. There is evidence that Pueblo farmers in the Northern Rio Grande used irrigation ditch technology prior to the Spanish-introduced acequia system to take advantage of this irrigable land (Davis, 2022). Indeed, the Cuyamungue Field School and Davis dissertation surveys documented two such ditches. Pueblo innovation in developing agricultural technologies, including ditch irrigation and gravel mulch fields, clearly helped to overcome environmental constraints.

Maize was the best crop to grow in the sense that the greatest proportion of area could support growing it, followed by wheat, then cotton. These differences could help explain the quicker adoption of sheep (for wool) versus the slower adoption of wheat, since maize was already a successful and culturally important crop whereas generally marginal environmental conditions for cotton meant that farmers had to invest significant additional energy to create microenvironments in which it could be grown. Sheep, in contrast, could live off of a wide range of vegetation that occurred naturally in uncultivated areas.

Expanding the model to include more precontact and contact villages across the Northern Rio Grande showed that the local environment was not a dominant factor behind the differential persistence of Pueblo villages during the contact period. Some villages that were depopulated during the contact period had higher percentages of their agricultural catchments in the crop growing niches than other villages that persisted. Examining the niche proportions for the four study localities against the proportions for sites across the Northern Rio Grande region demonstrated that the four study sites are representative of the agricultural environment of the region as a whole. Additionally, the agricultural potential model highlights the importance of having locally resilient varieties of crops. The growing conditions presented in the model, based on dendrochronology dating to the time period of interest and paleoclimate data, were more favorable than written historic accounts suggest due in large part to the locally adapted resilient varieties of plants being grown by these Pueblo communities.

Conclusion

The data presented in this paper provide niche reconstructions for three crops of importance to Ancestral Pueblo communities in the Northern Rio Grande region of New Mexico: maize, cotton, and wheat. The model suggests that the (passive) maize niche was relatively consistently met, followed by wheat, with cotton having a very limited niche that required additional agricultural technologies called gravel mulch fields. These results are supported by pollen and phytolith data and provide an additional line of evidence to support a slower adoption of wheat relative to maize versus a quicker adoption of sheep (wool) relative to continued cotton cultivation by Ancestral Pueblo communities in the Northern Rio Grande. The potential yield data showing agricultural surpluses in some villages provides an additional line of

evidence that agricultural products were part of the Ancestral Pueblo trade economy (Ortman (ed.) 2019). The model suggests that cotton growth in New Mexico in the protohistoric period was more constrained by limited Growing Degree Days than it was by limited precipitation. Thermal studies of gravel mulch fields demonstrate the utility of the gravels holding the heat into the evening, thus potentially staving off frost (Davis *et al.*, 2024). Finally, the model adds another data point to Southwest agricultural niche reconstruction efforts and extends the coverage of the PaleoCAR model to additional crops and regions of the Southwest.

Generalizing from the case study, we recommend that an agricultural potential model that incorporates cultural practices should have data on GDD and PPT for the area, agronomic variables for the crops of interest, archaeobotanical data for ground-truthing and adjusting the model based on where crops were actually growing, and data for environmental or anthropogenic features that align with the archaeobotanical locations. Then the resulting yield estimates can be compared against population estimates and the socioeconomic record (such as trade and taxation practices) to situate the modeled area in its greater cultural context.

While the concept of constructing and editing the model is broadly applicable to researchers interested in agricultural systems, their spatial extent, and their socioeconomic implications, environmental variation and local foodways/practices necessitate that the model be tailored to each case study/region (Bocinsky & Kohler, 2014). Understanding agricultural possibilities helps understand a region's settlement patterns and a community's actions. While set in the American Southwest, the crops in the case study to test the model (maize, wheat, and cotton) and the socioeconomic pressures (colonization and forced increased production through taxation) are relevant beyond the American Southwest (Logan *et al.*, 2024). The model may be adapted for the agronomic and paleoclimatic variables of another region. Comparing the yield estimates of the model, augmented by technological innovations to overcome environmental constraints to improve agricultural productivity (survey and archaeobotanical data), to the taxation requirements (historical data) and population estimates (archaeological data), demonstrated innovative agricultural solutions derived by ancestral people to respond to social and environmental pressures, supporting studies that observed similar patterns in other regions (De Varielles *et al.*, 2021).

The broader utility of this case study is that it illustrates agricultural intensification as a form of niche construction. To analyze archaeological niche construction, researchers need to understand where these practices (active modifications/agricultural aid feature construction) are happening on their landscape of interests and where the passive agricultural niche is. Only by overlaying these two things in a cultural context can one say something about where one might expect active cultivation techniques to exist and why. While niche construction theory is a common consideration in archaeological agricultural research, previous research has been limited by the challenges of factoring relative effectiveness or differential success of particular crops and agricultural techniques and the cultural interventions that people were employing to improve agricultural success at a regional system scale in a quantifiable way. If archaeologists are to interpret the suite of agricultural practices and socioeconomic and environmental pressures in an integrated way, such as

how pressures of agricultural product taxation in the encomienda system differentially impacted communities, then we have to have a sense of the baseline of the passive niche for those communities relative to their regional neighbors (whether a crop would likely flourish with limited investment or whether other niche construction techniques could be required). Quantifying the passive niche and cultural pressures helps contextualize and explain investment in intensification strategies, such as constructing gravel mulch fields, and explains the shift to and then away from this field construction practice. Customizable code templates with re-runnable functions, available via GitHub (https://github.com/bocinsky/paleocar), provide researchers with a tool to illustrate and quantify passive agricultural niches for their crops and locations of interest to help interpret the necessity of and help explain the adoption of technological practices in an environmental and cultural context.

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Author Contribution K.E.D wrote the main manuscript, ran the code written by R.K.B., and prepared the tables and figures. R.K.B. wrote the R script (code) that generated the data and figures. All authors reviewed the manuscript and assisted with revisions.

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Data Availability Due to the sensitivity of site location data, the raw data and detailed code may be obtained by request. Please email the corresponding author (Kaitlyn.Davis@nau.edu) to request access to the data and detailed code.

Declarations

Ethics Approval The research was conducted in accordance with ethical and best practice guidelines issued by the American Anthropological Association (AAA) Ethics Guidelines, the Society for American Archaeology (SAA), and the Register of Professional Archaeologists (RPA). Both authors areactive RPA members.

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